

NLP-based Psychological Diagnostic Diary

Midterm Presentation

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Progress

① UX/UI Design ② Front-end ③ Back-end ④ AI

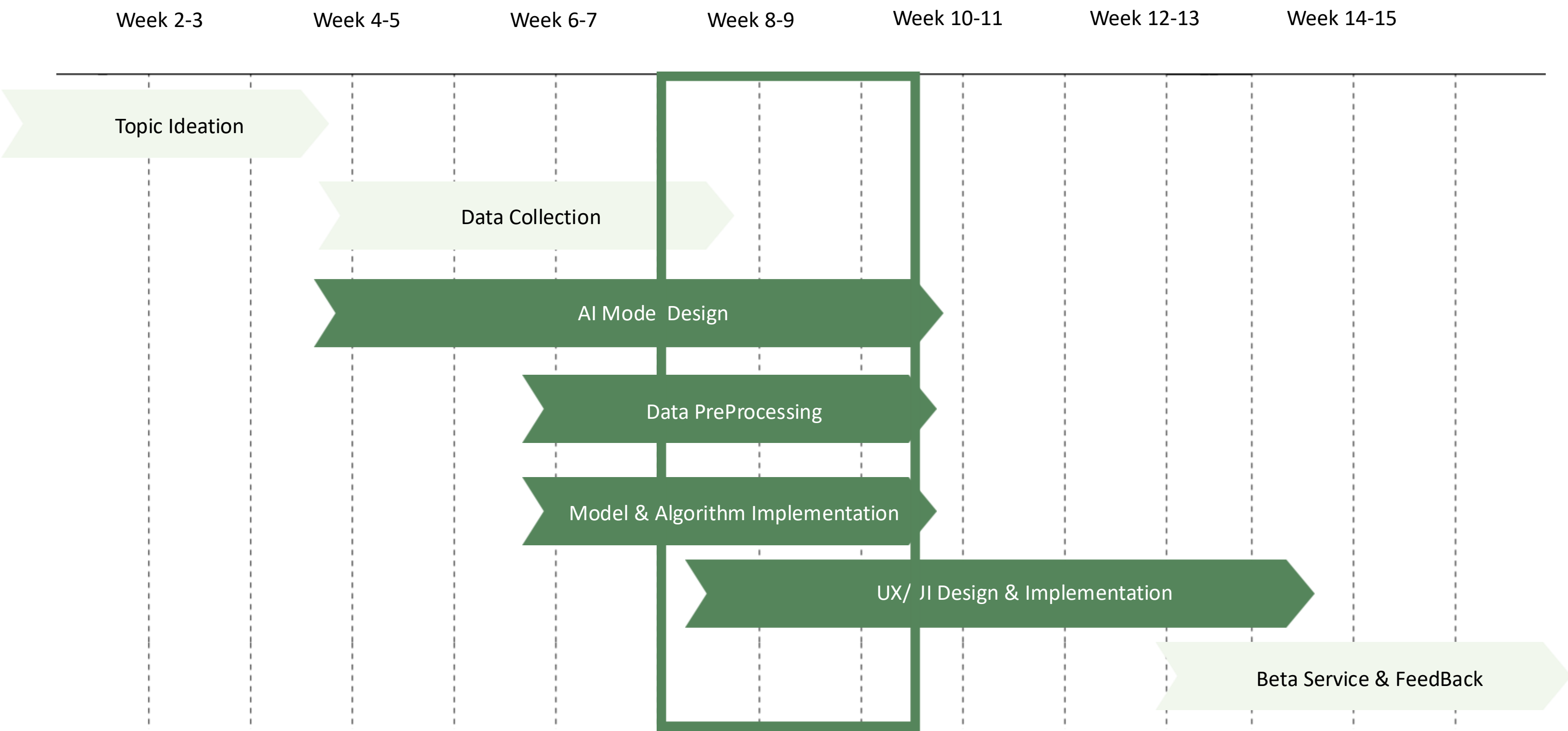
03

Challenges / Limitations

04

Plan

01. Planned Schedule



01. Role of Each Member



Yehdarm

UX / UI design



Nayeong

AI Implementation



Giho

Front-end

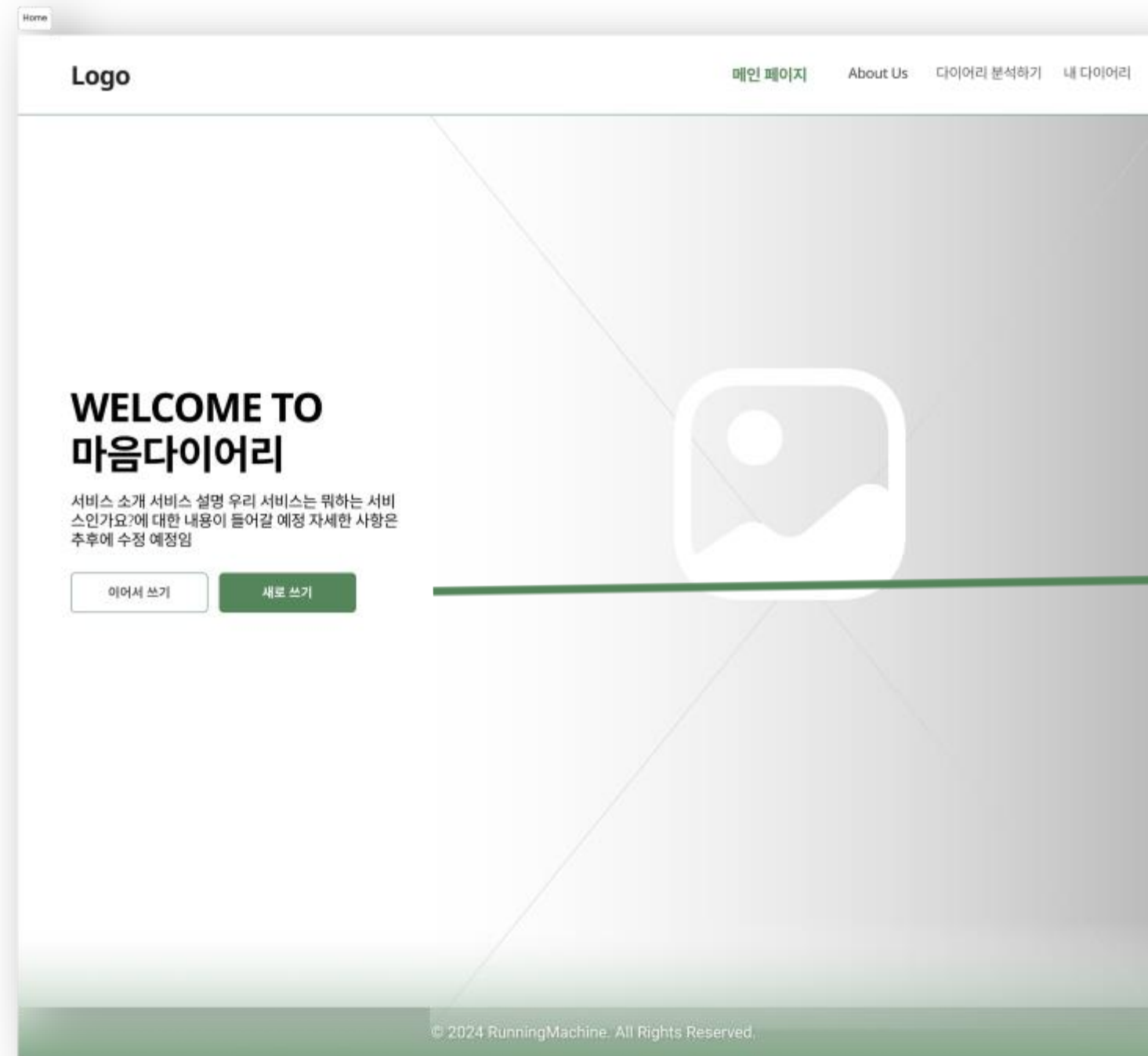


Junhyun

Back-end

02. Progress - UX / UI Design

<Main Page>



Navigation Bar

Button to Writing Page

02. Progress - UX / UI Design

<Writing Page>

일기 작성 페이지_새로 쓰기

✕ 새로 쓰기

날짜를 선택하세요

장소를 입력하세요

태그 입력

임시 저장 작성 완료

제목을 입력하세요.

B I U T B L I N E

내용을 작성하세요.

Create New Journal

일기 작성 페이지_이어서 쓰기

✕ 이어서 쓰기

2024. 10. 18. FRI

Capstone Class

태그 입력

임시 저장 작성 완료

Weekly Progress Meeting

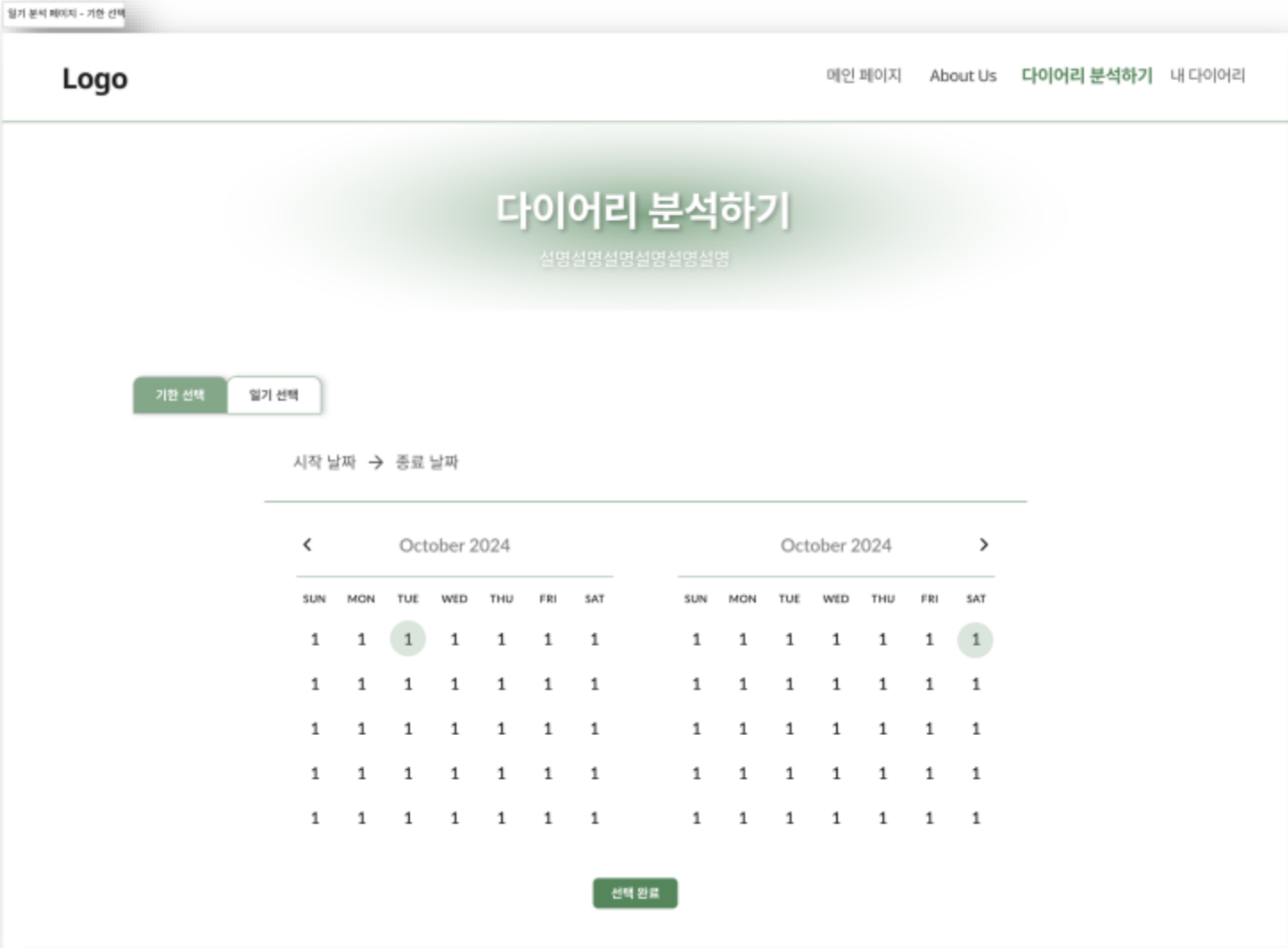
B I U T B L I N E

오늘은 캡스톤 수업의 위클리 발표날이었다.

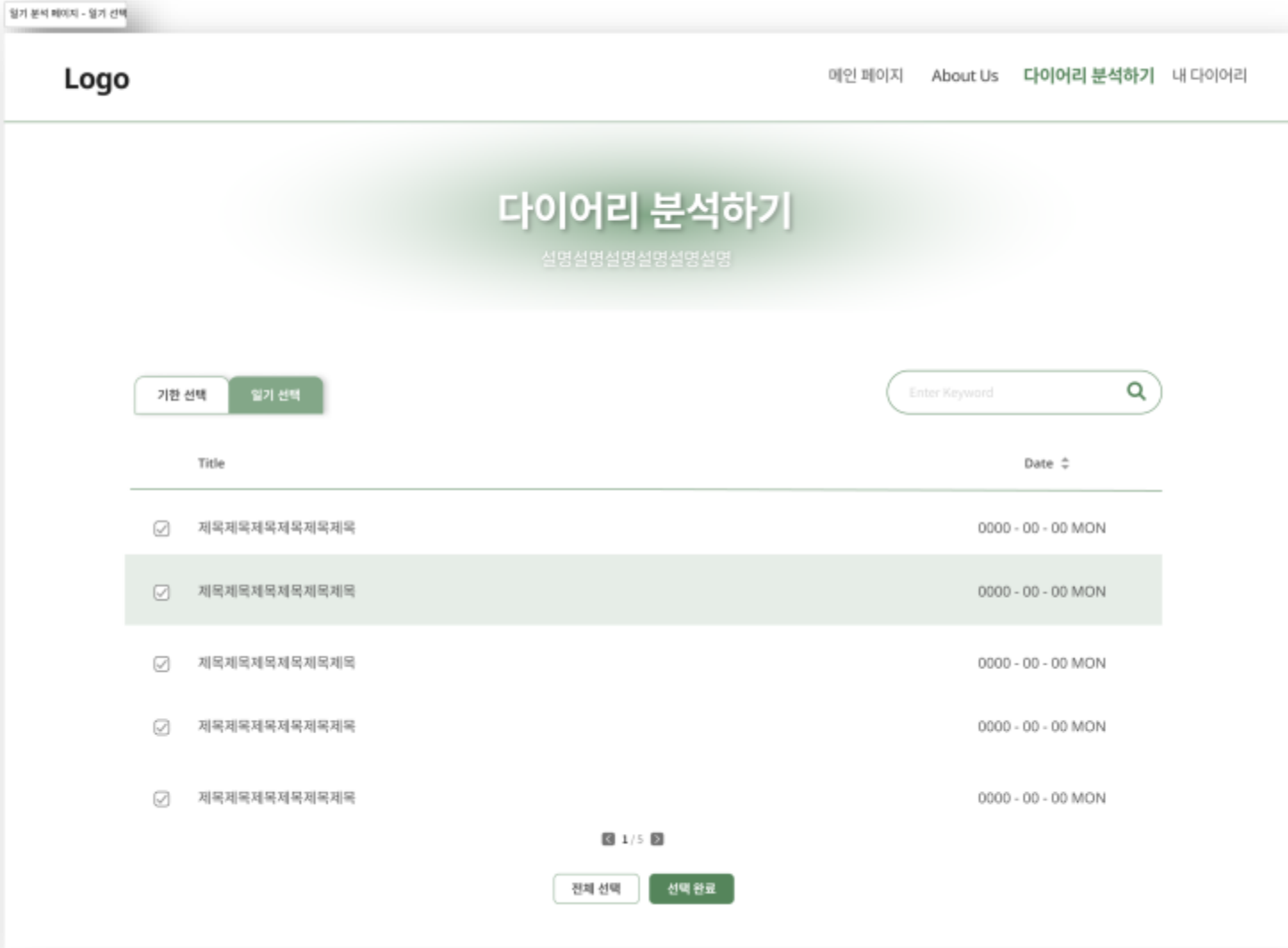
Use Drafted Journal

02. Progress - UX / UI Design

<Analyzing Page>



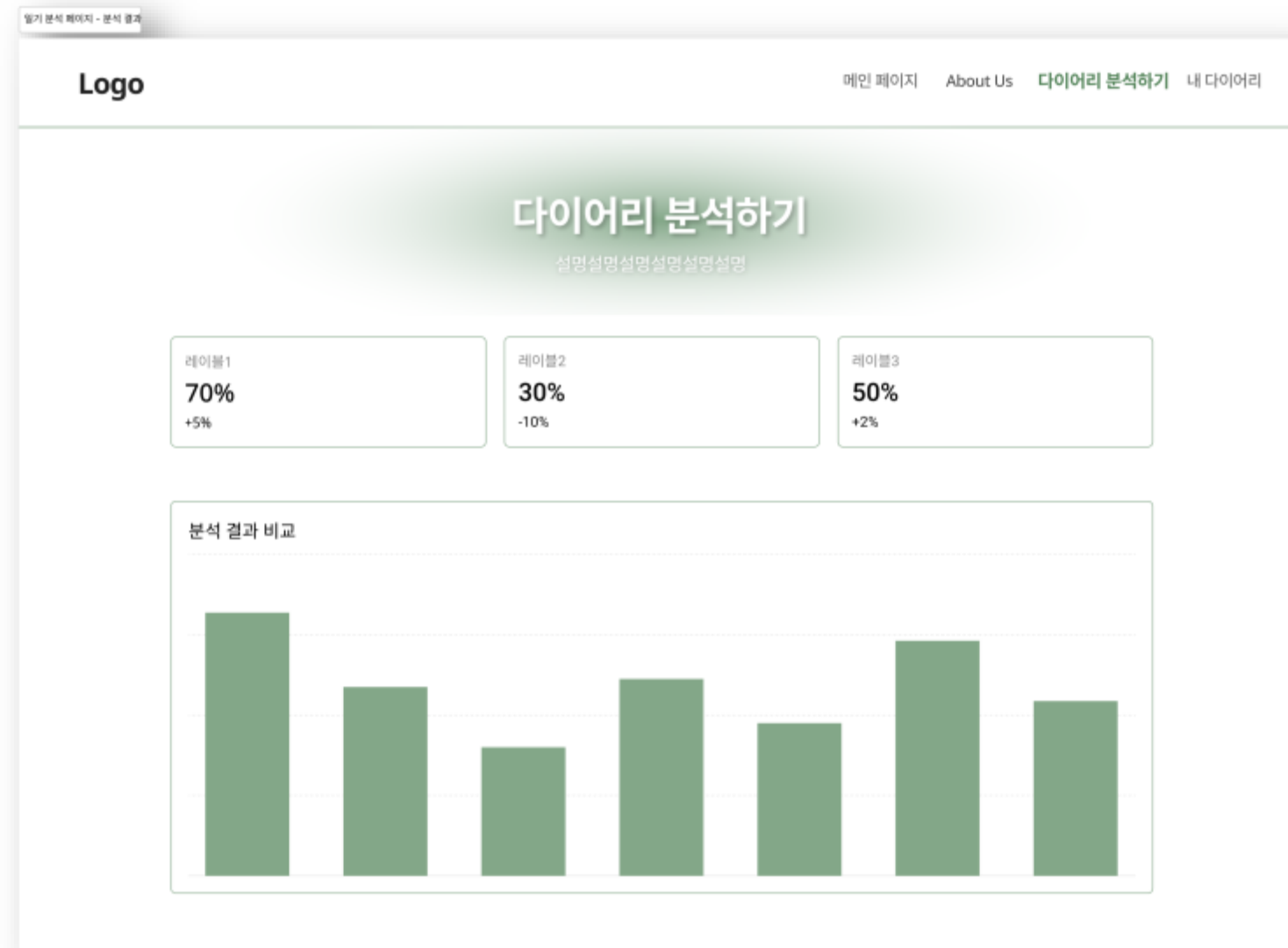
Choosing by Date



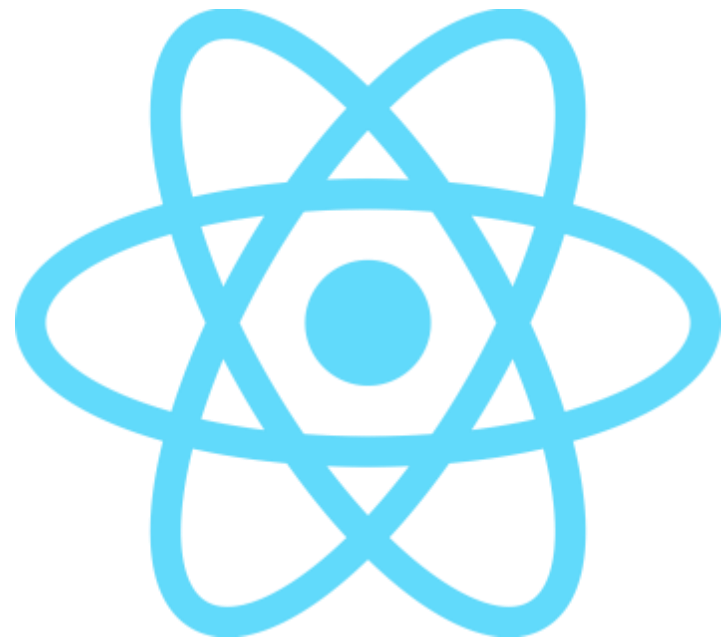
Choosing Manually

02. Progress - UX / UI Design

<Analyzing Page>



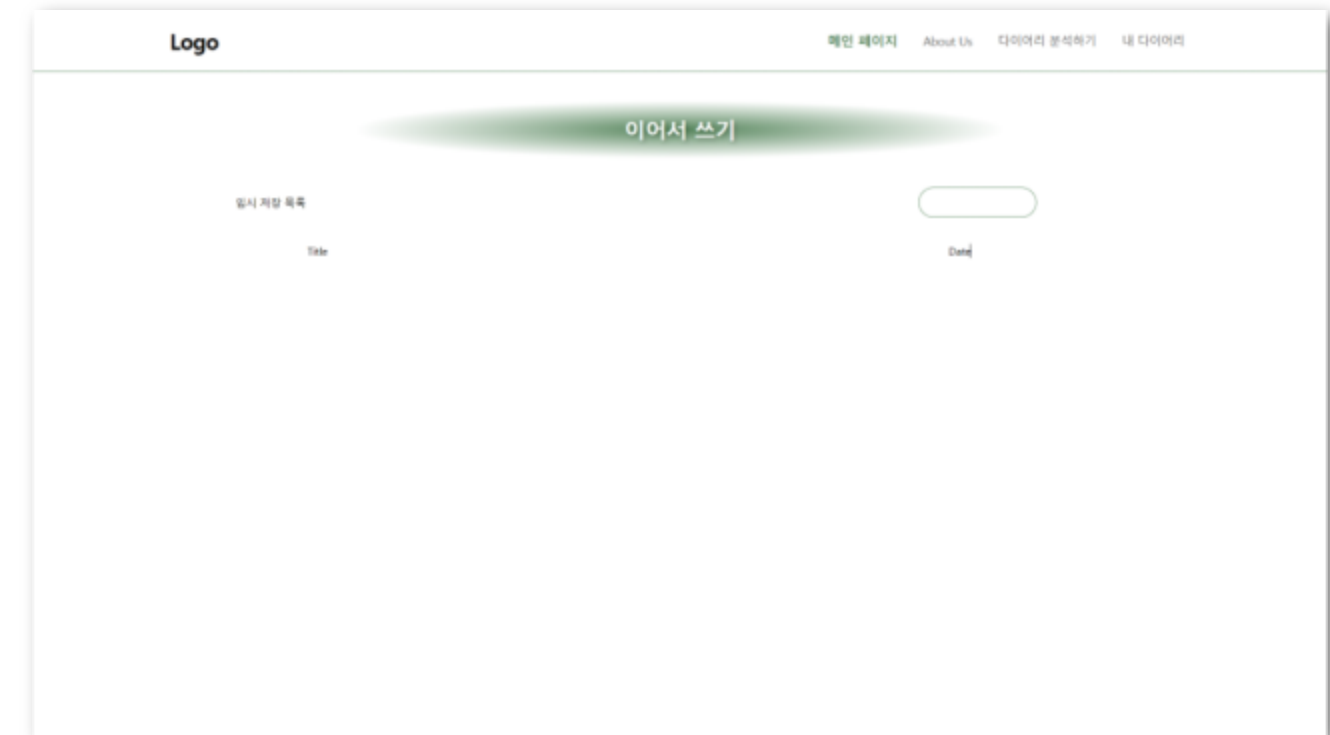
02. Progress – Front-end



Utilizing **React** to build Web pages

1. easy to **reuse** components
2. high performance in **changing** web pages

02. Progress – Front-end



02. Progress – Front-end



For communication with Back-end

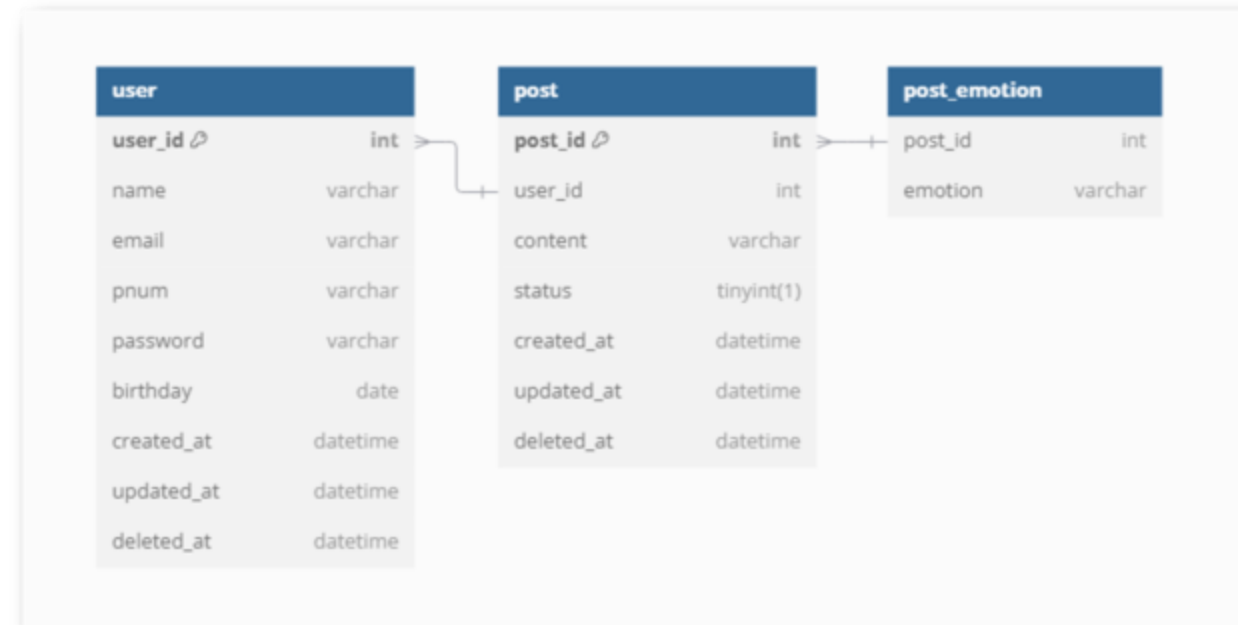
1. more **comfortable** and **intuitive**
2. provide more **secure** environment

02. Progress – Front-end

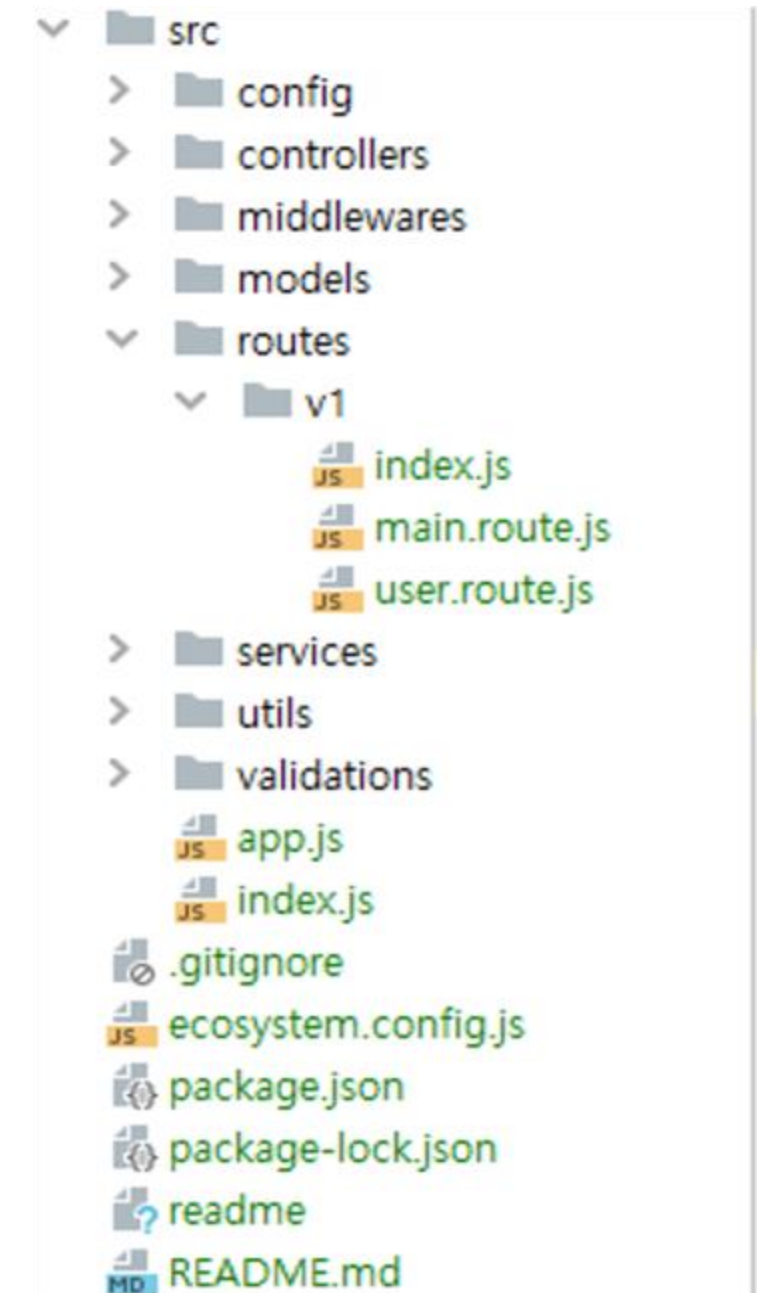
API List

user
GET get user
POST create user
POST login
main
GET getPostList
GET getPost
POST createPost
PUT editPost
DEL deletePost
GET getPostEmotions
GET getPostListEmotions
POST createPostEmotions

ERD



Framework



02. Datasets - AI

Data Collection



Wellness Conversation Script Dataset

From 16,000 **counsel data** received from Gangnam Severance,
5,232 user utterances were extracted
when the patient himself/herself visited the hospital.

Depression / Sadness / Loneliness / Anger / Lethargy / Abnormal Emotional Control
Sense of Loss / Loss of Appetite / Increase in Appetite / Insomnia
Nervousness / Fatigue / Feeling of Guilt / Loss of Concentration
Decrease in Confidence / Decline in Self-esteem
Feeling of Despair / Suicidal Impulse / Feeling of Insecurity

19 emotional labels in total

02. Model description - AI

SKTBrain/**KoBERT**

Korean BERT pre-trained cased (KoBERT)



5
Contributors



3
Issues



627
Stars



159
Forks



A version of BERT fine-tuned for Korean

02. Model description - AI

Why?

BERT

Bidirectional to understand context
back and forth

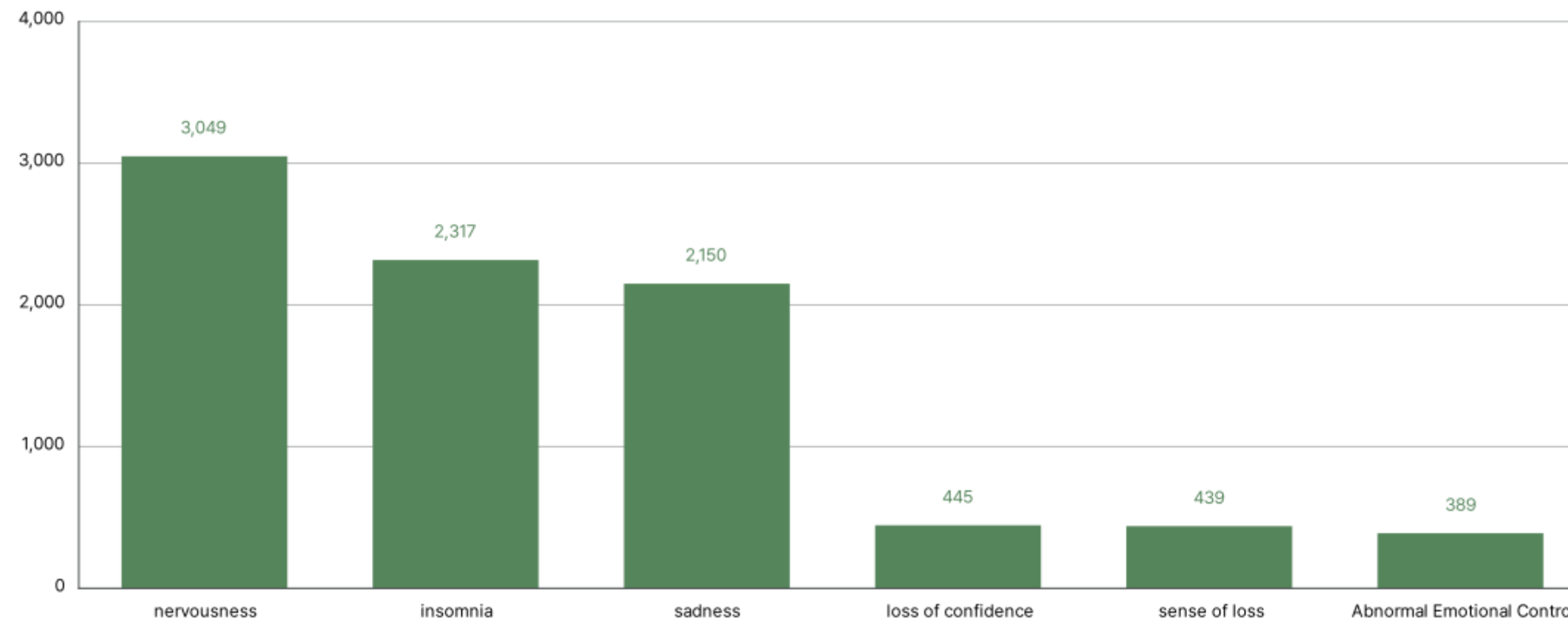


KoBERT

Optimized for Korean
reflecting irregular language changes in Korean

02. Data augmentation - AI

Imbalance in the number
of utterances by emotion label



02. Data augmentation - AI

nervous	3049
insomnia	2317
sadness	2150

...

loneliness	956
hopeless	887
suicidal	727
appetite↑	540
concentration↓	458
confidence↓	445
loss	439
emotion control↓	389

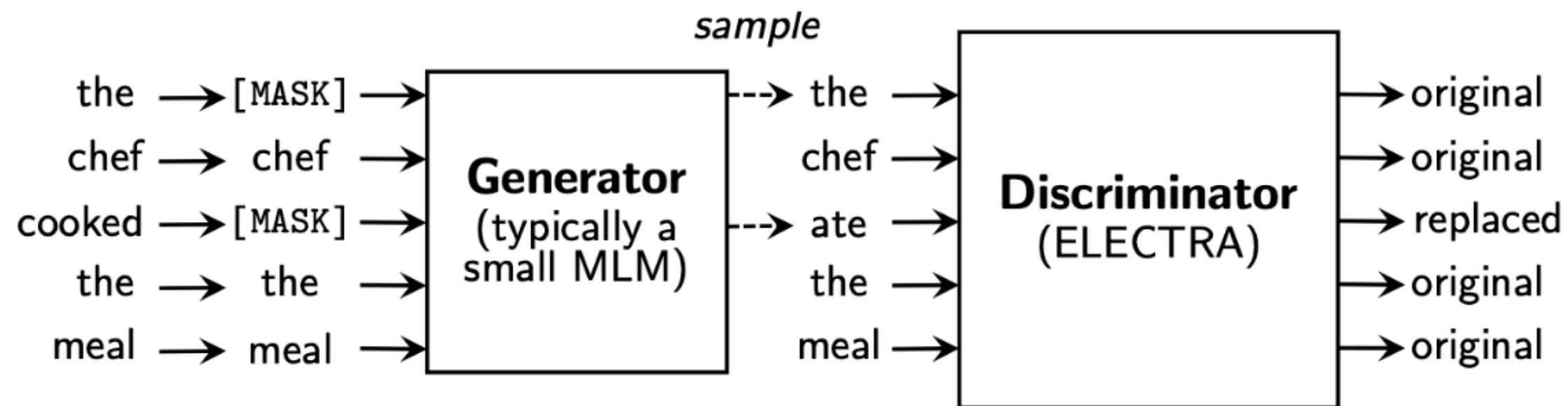


Data Augmentation
for 8 emotions
with less than 1000 utterances

02. Data augmentation - AI

< ELECTRA >

Efficiently Learning an Encoder that Classifies Token Replacements Accurately



Replaced Token Detection (RTD): Binary classification problem where **generator** turns some of the tokens into a plausible fake token and **discriminator** guesses whether each token is a **real** or a **replaced** token

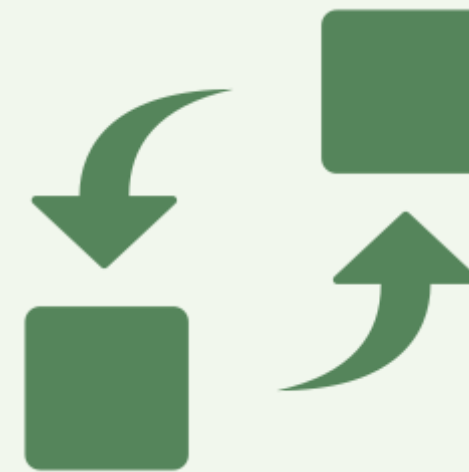
02. Data augmentation - AI

< KoELECTRA >

A natural language model that learns ELECTRA with **34GB of Korean text**



Korean-specialized



**Efficient learning
structure**

02. Data augmentation - AI

1. Random Masking Replacement

그런데 완전히 죽기에는 3층이 너무 낮았던 거죠.



Randomly

그런데 [MASK] 죽기에는 3층이 너무 낮았던 거죠.



KoELECTRA's fill-mask pipeline

그런데 **이게** 죽기에는 3층이 너무 낮았던 거죠.

02. Data augmentation - AI

2. Random Masking Insertion

그런데 완전히 죽기에는 3층이 너무 낮았던 거죠.



Randomly

그런데 완전히 죽기에는 3층이 너무 낮았던 **[MASK]** 거죠.



KoELECTRA's **fill-mask** pipeline

그런데 완전히 죽기에는 3층이 너무 낮았던 **그런** 거죠.

02. Data augmentation - AI

Random Masking Replacement



Vocabulary diversity
-> Different expressions
for context



Limited semantic change
-> Generates similar data

Random Masking Insertion



Various expressions
-> Can be transformed or
emphasized



Meaning distortion
-> Meaning of a sentence can be
distorted

02. Evaluation Metrics - AI

epoch = 5, batch size = 32, learning rate = 5e-5

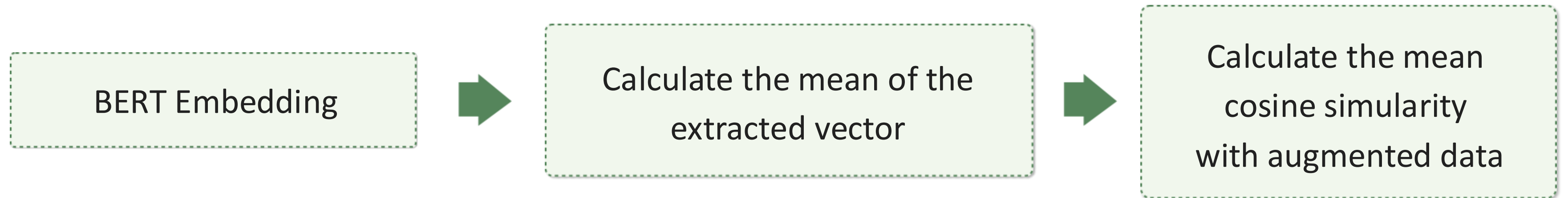
	Test Acc	F1-score	Precision	Recall	Cosine Similarity
base	0.840	0.871	0.861	0.863	-
Only Replacement	0.876	0.855	0.856	0.856	0.951
Only Insertion	0.867	0.881	0.864	0.885	0.973
Both	0.865	0.864	0.864	0.865	0.964

02. Evaluation Metrics - AI

< Cosine Similarity >

The **similarity** between two vectors that can be obtained using the **cosine angle** between the two vectors

$$similarity = \cos(\Theta) = \frac{A \cdot B}{||A|| ||B||} = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}}$$



03. Challenges - Front-end

< October 2024 >						
Su	Mo	Tu	We	Th	Fr	Sa
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2



시작 날짜 → 종료 날짜

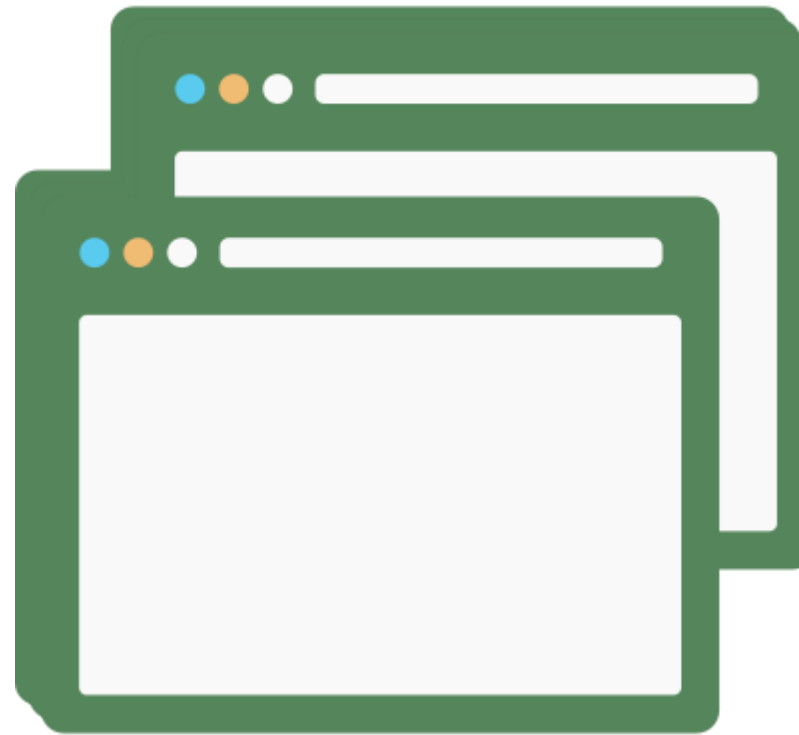
< October 2024							October 2024 >						
SUN	MON	TUE	WED	THU	FRI	SAT	SUN	MON	TUE	WED	THU	FRI	SAT
1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1

Limitations of the imported module, **datepicker**



customize it and make our own datepicker

03. Challenges - Front-end



Keep the page at the same time as the details change



Create a tab component

03. Challenges - Back-end

Handling Large-scale Data

Large Data

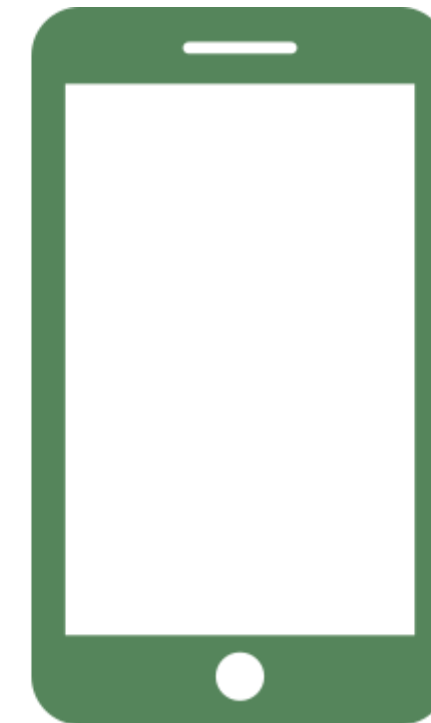
Make Slow



Optimized Database Indexing

Distributed Processing Technique

03. Limitations - Front-end



Limitations of fitting web pages based on **user's environment**

03. Limitations - Back-end

Potential Delay in Real-time Analysis

Export emotion analysis Results

Need constant time



Response Delay

04. Plan

UX/UI Design

Advance existing design

Create a Branding Design
Logo CI, Color Chip, Typography

Planning Additional Features and
Building Wireframes

Front-end

Implementation
of an Unimplemented Pages

Back-end

Developing Framework

AI

Hyperparameter tuning

Identify the KoELECTRA architecture and
devise ways
to improve performance

AI model serving with flask

Q & A

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